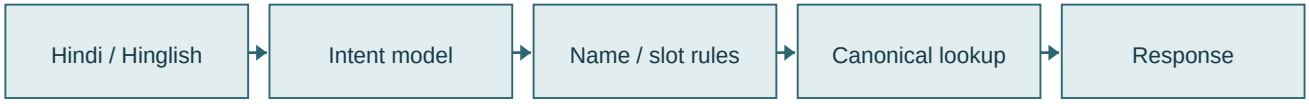
Accessible Hindi / Hinglish Transport Assistant Using NLP

ACADEMIC PROTOTYPE REPORT - 12 OCTOBER 2026

A limited Chennai case study that demonstrates intent classification, known-place entity extraction, a sourced transport lookup and Hindi/Hinglish responses. The interface exposes the NLP stages for explanation and error analysis.

Scope: a local nine-intent NLP demo. Four example station contexts and seven curated locations support demonstrations. Name recognition searches all original canonical station, bus-stop and landmark records and stored aliases; route/timetable coverage remains limited.



Intent	Requested information
Route	How to reach a destination; last-mile options
Availability	Whether a transport connection exists
Timing	Scheduled first/last departures or timetable
Station information	Station details and nearby bus stops
Accessibility	Lift, wheelchair and barrier-free facilities
Ticketing	Ticket, reservation, fare or payment
Train status	Current delay, position or cancellation
Railway complaint	Cleanliness, water, food, staff or coach problems
Out of scope	Non-transport questions

Delivered: one-process Streamlit demo, baseline/neural comparison, small catalog, generated-data evaluation and a demonstration guide. The prototype does not book tickets, file complaints or fetch live train status.

Generated data and development-exposed checks do not establish real-passenger accuracy. Human language review remains outstanding.

Data and NLP method

Dataset part	Rows	Origin
Template variants	1,452	Synthetic teaching examples
Short railway seeds	20	Synthetic development examples
Varied examples	180	AI-authored development inputs
Total intent data	1,652	1,090 Hindi / 562 Hinglish

Training	Validation	Test
1,112	216	324

Template families, including bilingual variants and place substitutions, stay within one split. AI-authored development pairs are training-only and replace repetitive rows. The linked slot dataset has 320 rows, including 72 generated test queries.

TF-IDF baseline: character n-grams of lengths 2-5, sublinear frequency and Logistic Regression (C=4, max_iter=1000, seed 42). It remains the default because its validation macro F1 is stronger.

MuRIL experiment: reuse the existing transport-trained checkpoint; adapt only the last two encoder layers and a new nine-class linear head. The embeddings and earlier layers stay frozen. Attention-masked mean pooling supplies the representation.

Setting	Value
Epoch budget / selected epoch	5 / 1 (validation macro F1 only)
Batch / maximum tokenizer length	16 / 64 tokens
Optimizer	AdamW; weight decay 0.01; clipping 1.0
Learning rates	Encoder 0.00005; head 0.001
Input / seed	Raw query / 42

Two data versions were trained with the same settings after qualitative failures exposed narrow route wording and missing scheduled-rail examples. Exact check questions were excluded from training. This is partial fine-tuning of reused weights, not a fresh pretrained-model benchmark.

Entity extraction: known-name gazetteer plus Hindi/Hinglish role, mode and feature rules. There is no learned NER model. Rules do not override the displayed classifier intent.

Measured results and errors

Generated template-family test partition

Model	Language	Rows	Accuracy	Macro F1
tfidf	hi	216	96.3%	0.962
tfidf	hinglish	108	100.0%	1.000
muril_finetuned	hi	216	83.3%	0.832
muril_finetuned	hinglish	108	69.4%	0.660

Validation macro F1: baseline 0.981; MuRIL 0.882. The baseline remains stronger on the declared partitions.

AI-authored development diagnostic

Model	Intent accuracy	Macro F1	Pipeline success
tfidf	95.6%	0.955	93.3%
muril_finetuned	82.2%	0.791	80.0%

Ninety questions in 45 bilingual scenarios check intent, supported slots, response status and exact catalog record IDs. Source attribution is required for returned records. The checks were development-exposed; they do not form a blind human benchmark.

Slot F1 is 1.000 on these generated checks, mainly covering curated example locations rather than all database names or general NER. Original status expectations are retained: the broader lookup can clarify ambiguity where the earlier whitelist returned unavailable. These scores do not establish passenger accuracy.

Baseline confusion matrix: generated test

True / pred	Route	Avail	Time	Station	Access	Ticket	Status	Compl	Other
Route	36	0	0	0	0	0	0	0	0
Avail	0	36	0	0	0	0	0	0	0
Time	0	8	28	0	0	0	0	0	0
Station	0	0	0	36	0	0	0	0	0
Access	0	0	0	0	36	0	0	0	0
Ticket	0	0	0	0	0	36	0	0	0
Status	0	0	0	0	0	0	36	0	0
Compl	0	0	0	0	0	0	0	36	0
Other	0	0	0	0	0	0	0	0	36

Remaining errors include scheduled timing versus train delay, and non-transport requests with travel-like wording. MuRIL also confuses some coach-facility complaints and out-of-scope inputs. Complete errors and neural confusion matrices are retained in the repository reports.

Demonstration and the user questions

Both final models classify the four user-supplied intents correctly. In the first pass the baseline classified only one correctly. General training-data corrections improved these development checks; they are not unseen accuracy evidence.

User question	Final baseline outcome
Guindy se Central kaise jayein?	route_query / information
Velachery se Park station kaise jayein?	route_query / clarification
Chennai Central se Mambalam station kaise jau?	route_query / clarification
Subah sabse pehle train guindy se chennai central kab hai?	service_timing / unavailable

Guindy to Central receives a published Blue Line Metro connection. Velachery, Park and Mambalam are recognized from the original database. Park/Park Town and physical-stop variants need clarification. Explicit railway names resolve, but railway route sequences and departure schedules are absent from this snapshot.

The original theme and fonts are preserved. Example selection, a new-question reset, readable slots, explicit answer status and same-question model comparison support a short demonstration.

Five-minute demo sequence

1. Enter a Hindi Central-to-Airport Metro availability question; show intent, slots, record and source.

2. Switch the saved response to Hinglish; the raw question and prediction remain unchanged.

3. Ask for Tambaram-to-Guindy bus availability; show the 51A stop-sequence snapshot.

4. Demonstrate Central Metro lift information, a dirty-toilet complaint and a late-train query.

5. Show an ambiguous station name or missing railway route and the evaluation/model comparison.

Catalog: eight Metro links, two bus connections, Central Metro accessibility and one historical terminal window. Supplementary Central/Velachery guidance names Park Town MRTS separately. Nearby-stop lookup separates anchor from target mode; official Guindy stop names can be returned without unreliable distances. These are source records, not live operation or verified walking transfers.

Timings: the terminal-window illustration is explicitly dated 16 October 2024. It cannot answer a current, destination-specific or railway departure. The official current timetable is linked for confirmation.

Reproduction, limits and references

Run the local demo

```
.venv/bin/python -m streamlit run app/streamlit_app.py \
--server.address=127.0.0.1 --server.port=18505
```

Open http://127.0.0.1:18505. Install the README dependencies first. The baseline is included in Git; MuRIL uses the verified release download. One Streamlit process is sufficient.

Reproduce the experiment

```
python -m scripts.prepare_models
python -m scripts.nlp_v2.build_project_dataset
python -m scripts.nlp_v2.train_project
python -m scripts.nlp_v2.evaluate_project
python -m pytest -q
```

The submitted runtime includes the baseline, nine-class head, pinned tokenizer and canonical database. Large encoder assets are distributed through a versioned release and verified during setup. Inference uses local files and needs no private checkpoint or Hugging Face cache. The exported encoder produced bit-for-bit identical features on 94 diagnostic/user questions. Shared lookup regressions remain in the submission; the retired API and earlier classifier are removed. Follow the README for a fresh Python 3.12 environment and model setup.

Limits and bounded follow-up

Generated text lacks human annotation review. Template grouping does not remove shared vocabulary or syntax. The checkpoint has prior transport training, and the diagnostics informed development. Name recognition searches the full canonical snapshot, including duplicates. It is not complete Chennai coverage: rail routes/schedules and a verified walking network are absent. Unlisted spellings, intent wording, complex negation and multi-intent requests can fail. Confidence scores are uncalibrated. The optional five-phrase Tamil lookup needs speaker review; speech and general translation are not implemented.

The next academic improvement is a small human-reviewed corpus and fresh natural evaluation, followed by analysis of the retained failures. The current deliverable is a working local prototype and a transparent experiment.

References

MuRIL: Multilingual Representations for Indian Languages

Scikit-learn TF-IDF vectorizer documentation

CMRL Metro line map

Central Metro station information

CMRL publication dated 16 October 2024

CMRL current timetable page

ChennaiGTFS community feed

Scope also follows the user-supplied original project description. Local source records, manifests, detailed metrics and the demo guide are included in the repository.